Date. ________ NO. ________

Radioactivity

Types: **Alpha radiation** - a release of particles, one of which is the alpha particle
"α"

X ↗ α ↘ Y

↳ $^{4}_{2}He$ is the only alpha particle released

Beta radiation - a release of particles, one of which is the beta particle
"β"

X ↗ β ↘ Y

↳ $^{0}_{-1}e$, electron
↳ $^{0}_{+1}e$, anti-electron/positron

} Possible beta particles released

positron ← antimatter of electron ← we (as matter) can't see antimatter

Gamma radiation - a release of gamma electromagnetic waves (high energy photons)
"γ"

X ↗ γ ↘ Y

↳ highest frequency waves (come from nucleus)

All decays are random and spontaneous

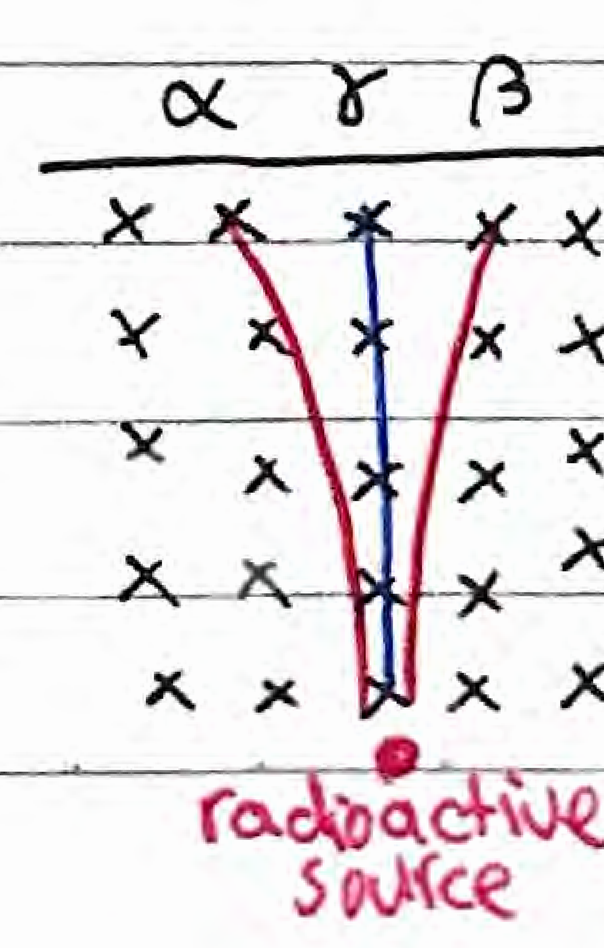
random ↓ in a sample of species, any particle can decay

spontaneous ↓ can happen any time

↳ These decays follow conservation of energy

Distinguishing types of radiation

- You can distinguish radiation due to charge using a magnetic field
  - ↳ Gamma (neutral, no deflection)
  - ↳ Alpha & Beta-positron (positive, left)
  - ↳ Beta-electron (negative, right)

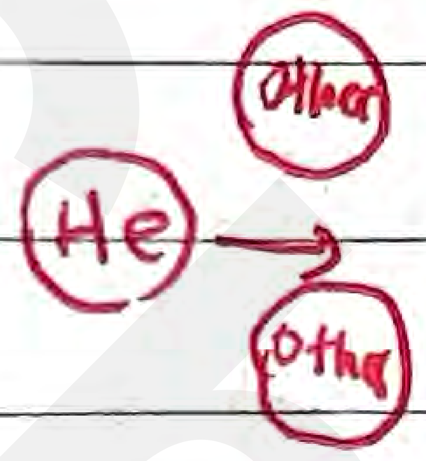


Date. ________ NO. ________

Radioactivity

Alpha particles:
- Highly ionizing (due to its 2+ charge)
- Penetrate matter poorly (large size due to being an ion)

same size magnitude as other atoms



- Are absorbed by a few cm of air or thin sheet of paper

Beta-minus particles:
- Moderate ionizing power compared with alpha particles of the same energy
- Moderate ability to penetrate matter
- Absorbed by ≈25cm air
- Absorbed by few cm of animal/plant tissue
- Absorbed by few mm of aluminum

Gamma photons:
- Weakly ionizing
- Eject electrons from metals (photoelectric effect), causing secondary ionization
- Highly penetrating
- Absorbed by few cm of lead
- Absorbed by few m of concrete
- Only weakly absorbed by animal tissue

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Alpha Decay

$${}^{A}_{Z}X \longrightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$$

find element Y from atomic #

E.g. ${}^{214}_{84}Po \rightarrow {}^{210}_{82}Pb + {}^{4}_{2}He$

Beta-minus Decay

occurs in neutron-rich nuclides (ratio between $\frac{\text{neutrons}}{\text{protons}}$)

tries to decrease neutron-richness

$${}^{A}_{Z}X \longrightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}e + {}^{0}_{0}\bar{\nu}$$

$\bar{a}$ denotes anti-

anti-neutrino (energy difference between product and reaction)

↳ To satisfy energy conservation

where did proton come from?

↳ protons are not the smallest indivisable particle

↳ neutron → proton + ${}_{-1}e$

made of quarks

E.g. ${}^{231}_{90}Th \rightarrow {}^{231}_{91}Pa + {}^{0}_{-1}e + \bar{\nu}$

Beta-plus Decay

occurs in proton-rich nuclides

tries to decrease proton-richness

$${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{+1}e + {}^{0}_{0}\nu$$

← neutrino (to balance energy)

proton → neutron + ${}^{0}_{+1}e$

E.g. ${}^{23}_{12}Mg \rightarrow {}^{23}_{11}Na + {}^{0}_{+1}e + {}^{0}_{0}\nu$

Electron-Capture Nuclear Reaction

- An alternative way in which proton-rich nuclides can increase the neutron/proton ratio. "artificial transmutation"
- Requires external help (i.e. for electron to hit nucleus)

E.g. ${}^{37}_{18}A + {}^{0}_{-1}e \longrightarrow {}^{37}_{17}Cl + \nu$ ← neutrino to balance energy

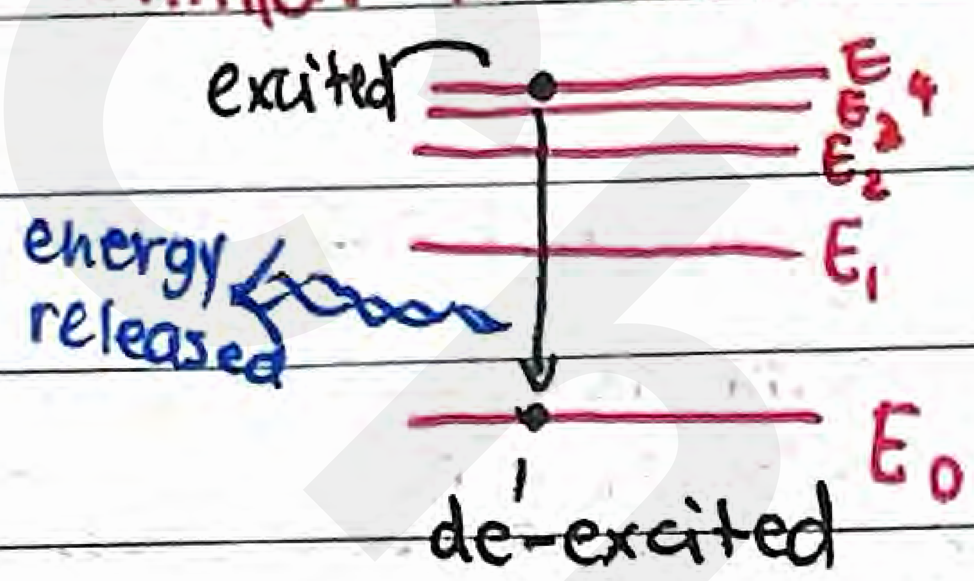
Date. ________ NO. ________

Gamma Decay

$${}^{A}_{Z}X^{*} \longrightarrow {}^{A}_{Z}X + {}^{0}_{0}\gamma$$

parent excited nucleus — daughter de-excited nucleus — gamma ray

Similar to electron energy levels, but for nucleus



will be said in question

↳ Typically follows from alpha/beta decays

↳ The product can sometimes be excited by which its unstable, thus gamma decay

The Radioactive Decay Law

- The rate of decay is proportional to the number of nuclei present that have not decayed yet.

rate of decay $= \dfrac{dN}{dt} \propto -N$ = # of undecayed neutrons

(negative since rate of change of undecayed nuclei)

$\underbrace{\dfrac{dN}{dt} \propto -N}_{\text{logistic differential equation}}$

- The more undecayed neutrons, the greater the rate of decay

$$\boxed{\frac{dN}{dt} = -\lambda N}$$

λ ← decay constant

↑ Per each isotope

The Radioactive Decay Law

Solving the differential equation...

$$\int_{N_0}^{N} \frac{dn}{n} = -\lambda \int_0^t dt_x$$

↳ initial (total nuclides)

$$\ln(n)\Big|_{N_0}^{N} = -\lambda t\Big|_0^t \Rightarrow \ln\left(\frac{N}{N_0}\right) = -\lambda t$$

Thus...

$$>>> N = N_0 e^{-\lambda t}$$

# of nuclei at a given time ← that haven't decayed yet

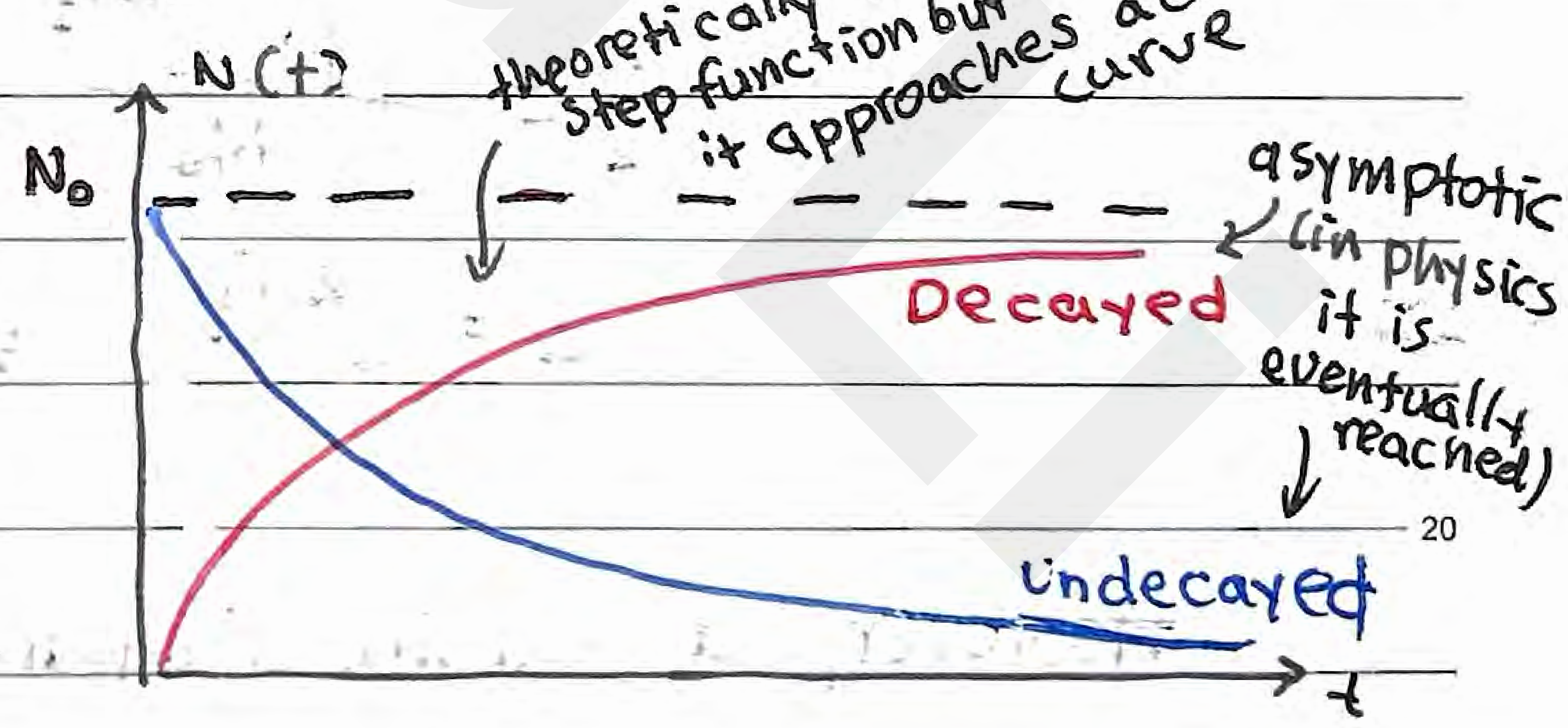
But how about # of nuclei that have decayed?

$$N_0 - N = \text{Total} - \text{Undecayed} = \text{Decayed}$$

$$\therefore \text{Decayed} = N_0 - N_0 e^{-\lambda t} = N_0(1 - e^{-\lambda t})$$

Thus (on a graph)...

- Decayed $= N_0 - N_0 e^{-\lambda t}$
- Undecayed $= N_0 e^{-\lambda t}$



Another word for the rate of decay is Activity

$$A = -\frac{dN}{dt} = \lambda N = \underbrace{\lambda N_0}_{\text{group}} e^{-\lambda t} = A_0 e^{-\lambda t}$$

initial activity

The unit is # of nuclei per second or Becquerel (Bq)

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Radioactive decay

The decay constant is constant for each isotope:

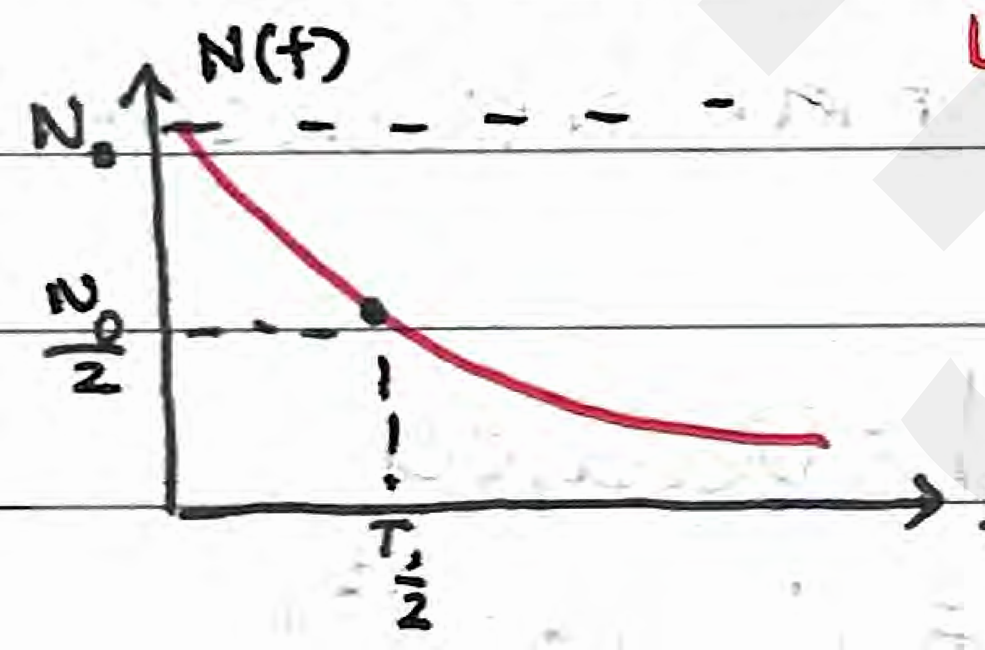
$$\lambda = \frac{\text{probability of decay}}{dt}$$

This constant connects with half-life

$$T_{1/2} = \frac{\ln 2}{\lambda} \approx \frac{0.693}{\lambda}$$

↳ How much time has passed for half of the initial nuclei (N_0) to decay?

↳ It is also time for Activity to half ($A \propto N$)



→ $2T_{\frac{1}{2}}$ is when N_0 decays to $\frac{N_0}{4}$

$3T_{\frac{1}{2}}$ is when N_0 decays to $\frac{N_0}{8}$

(keep halving)

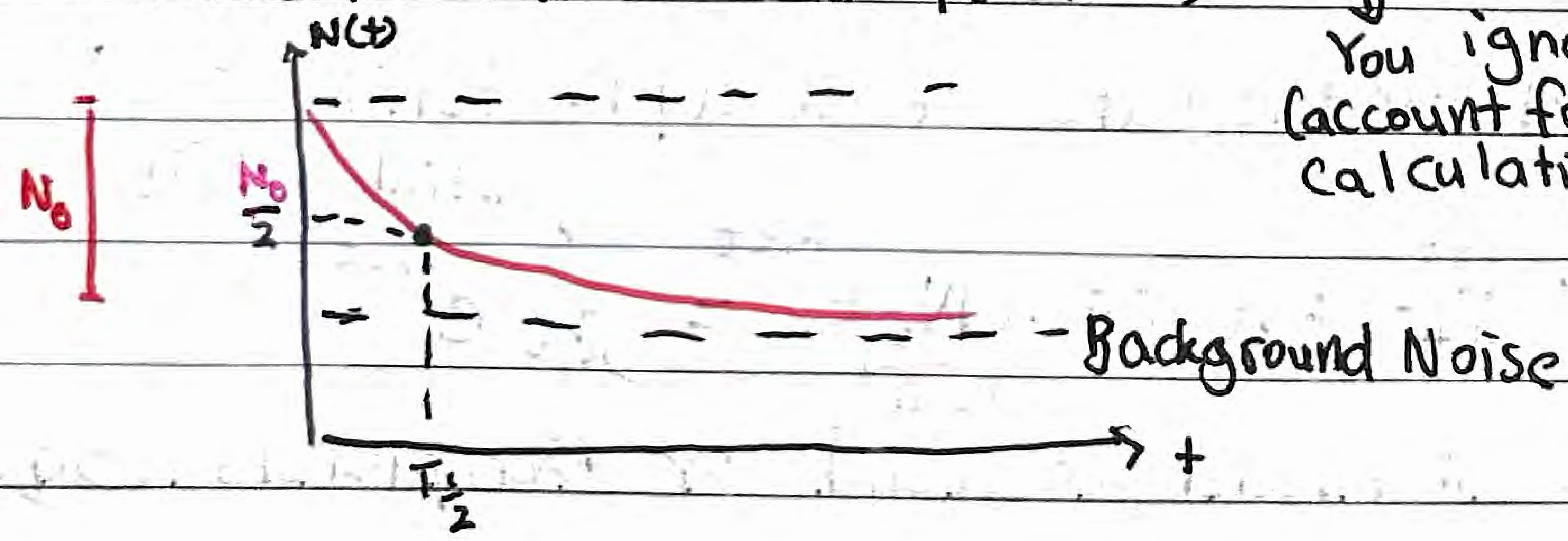
$$\frac{1}{2} = \frac{\cancel{N_0} e^{-\lambda T_{\frac{1}{2}}}}{\cancel{N_0}}$$

$$2 = e^{\lambda T_{\frac{1}{2}}} \Rightarrow T_{\frac{1}{2}} = \frac{\ln 2}{\lambda}$$

↳ useful to determine λ from half-life

However, a real-life measured decay graph has background noise too (unideal factors)

You ignore that (account for it) in calculating half-life



Radioactive Decay

- For time periods, typically there is no need for a conversion of units
  - ↳ When given ($T_{\frac{1}{2}}$) is mins, then your result (t) is also mins
- $\lambda \cdot N_0 = A_0$ (often expressed as decay rate before event observed)

Background Radiation (Noise)

- When a radiation detector is left without a prominent source of a radiation, it will still detect ionizing radiation
- The most typical type of detector is Geiger counter
- When a problem asks for final count after time, make sure to subtract background radiation, calculate, and add back the background radiation

Nuclear Reactions

- Transmutation - Transformation of one element into another
  - Can be both natural and artificial

↳ Occurs when a nucleus is struck by a nucleus, neutron or γ ray

↳ Reaction E.g.

target assumed to be motionless

$$n^0 + {}^{14}_{7}N \rightarrow {}^{14}_{6}C + p^+$$

neutron $\vec{v}_n$ + Nucleus → Atom $\vec{v}_A$ + proton $\vec{v}_p$

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Nuclear Reactions

- These reactions must follow conservation of particles
  - ↳ where $\#_{init} = \#_{final}$; $\#_{i,n^0} + \#_{i,p^+} \rightarrow \#_{f,n^0} + \#_{f,p^+}$
  - ↳ Thus, # nucleons is conserved, # protons is conserved (# neutrons is implied) ↳ rel. atomic mass

Reaction Energy (reactions must have energy)

- Since mass conservation is not guaranteed even with same total # of protons and neutrons, we can use mass difference for energy change

For general... $a + X \cdots \rightarrow Y + b \cdots$

- $Q = \Delta E$, $Q = (BE_Y + BE_{b\cdots}) - (BE_a + BE_{X\cdots}) = \Delta \Sigma BE$

alternatively (via mass difference...)

$E = mc^2 \rightarrow Q = -\Delta m \cdot c^2$ ← generalized for more terms

$= [(m_a + m_{X\cdots}) - (m_Y + m_{b\cdots})] c^2$

↳ Typically, mass difference is used to calculate binding energy

Just as in regular reactions... $Q > 0$, exothermic (energy released)

↳ mass decrease

$Q < 0$, endothermic (energy absorbed)

↳ mass increase

Nuclear chain reactions

↳ A particle launched will break nucleus, release more launched particles that initiate more reactions (breaking nucleus releases energy)

Nuclear Reactions

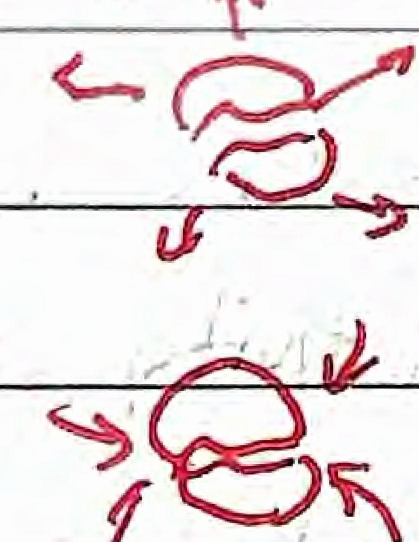
E.g. Uranium

$n \rightarrow {}^{235}U$ → Kr(A), Ba(B), n n n → continue on in a chain

For initial success, n must collide perfectly at center of mass ${}^{235}U$

After that (depending on speed of n_{init}), A and B are formed (not necessarily same element in every scenario)

large nuclei is decreasing segment of BE



Fission - Large nuclei split apart

↳ exothermic : Binding Energy, $BE_f > BE_i$

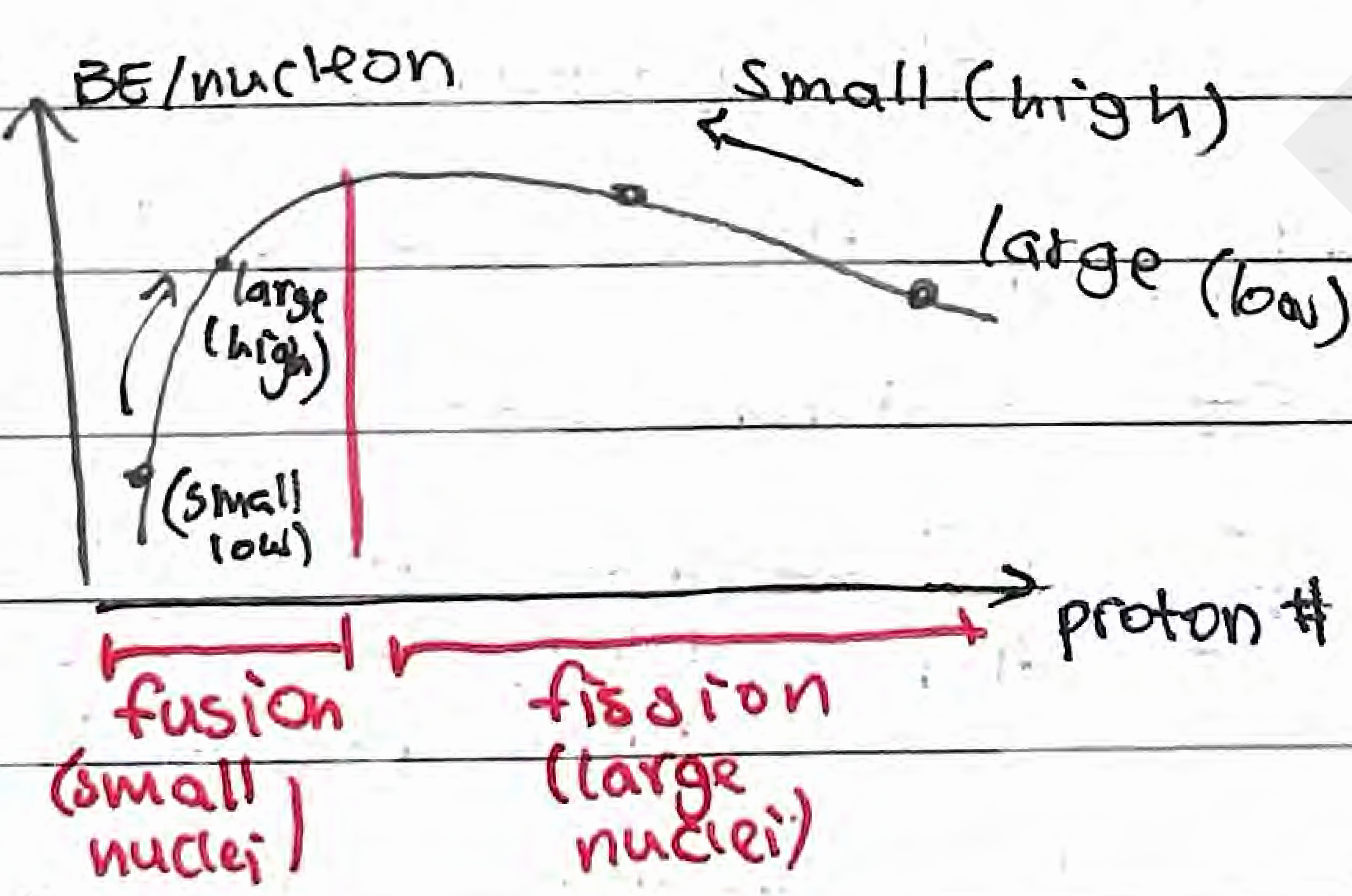
Fusion - Small nuclei join together → small nuclei, increasing segment of BE

↳ exothermic: , $BE_f > BE_i$

*How do we know if a reaction starts?

- Find Q of reaction, does initial supply of Q meet that? → To start, it's endothermic

Binding energy Fission & Fusion



Both fission and fusion are exothermic
↳ This is because BE depends on nuclei size

Additionally, heat can be calculated from binding energy graph
↳ Make sure to multiply by # of nucleons!!

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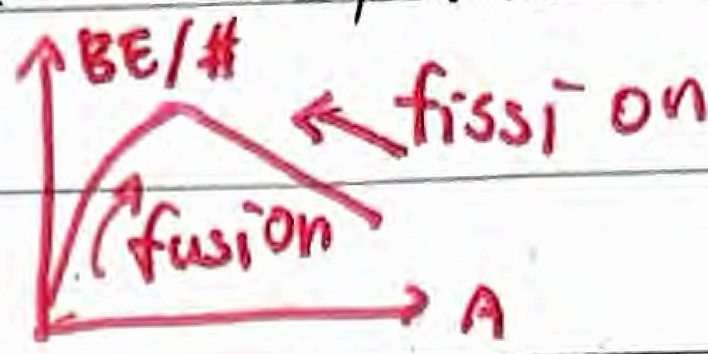
Fission - Process in which a heavy nucleus splits up into lighter nuclei

both are exothermic



Fusion - The joining of two light nuclei into a heavier one (can occur naturally or induced)

Radioactivity - Emission of particles and energy from a nucleus

↳ Random and spontaneous

Radioactivity and mechanics

↳ Radioactivity connects to momentum and energy

↳ Binding energy is typically ΔE_{tot}.

↳ This is reflected by the ΔE_k (when rest particle decays)

E.g. when $^{226}Ra \rightarrow ^{222}Rn + ^4_2He$, the $BE = \Delta E_{k\,tot}$.

When finding E_k, it's useful to observe momentum

↳ $\sum p_{int} = \sum p_{final}$, since $p = m \cdot v$, we can get E_K in terms of p and m...

$$E_k = \frac{p^2}{2m}$$

← The ratio between E_k of two particles is the ratio between their masses if $|P| = |P|$

↳ This is useful to substitute and simplify ΔE in terms of only one E_k, making $BE = \Delta E_k$ solvable

Radioactivity

- Additionally, remember binding energy is for one reaction — one nucleus
  - ↳ Thus, when given a sample, make sure to obtain how many nuclei there is and multiply for total energy.

$E=mc^2$ is $E \propto m$

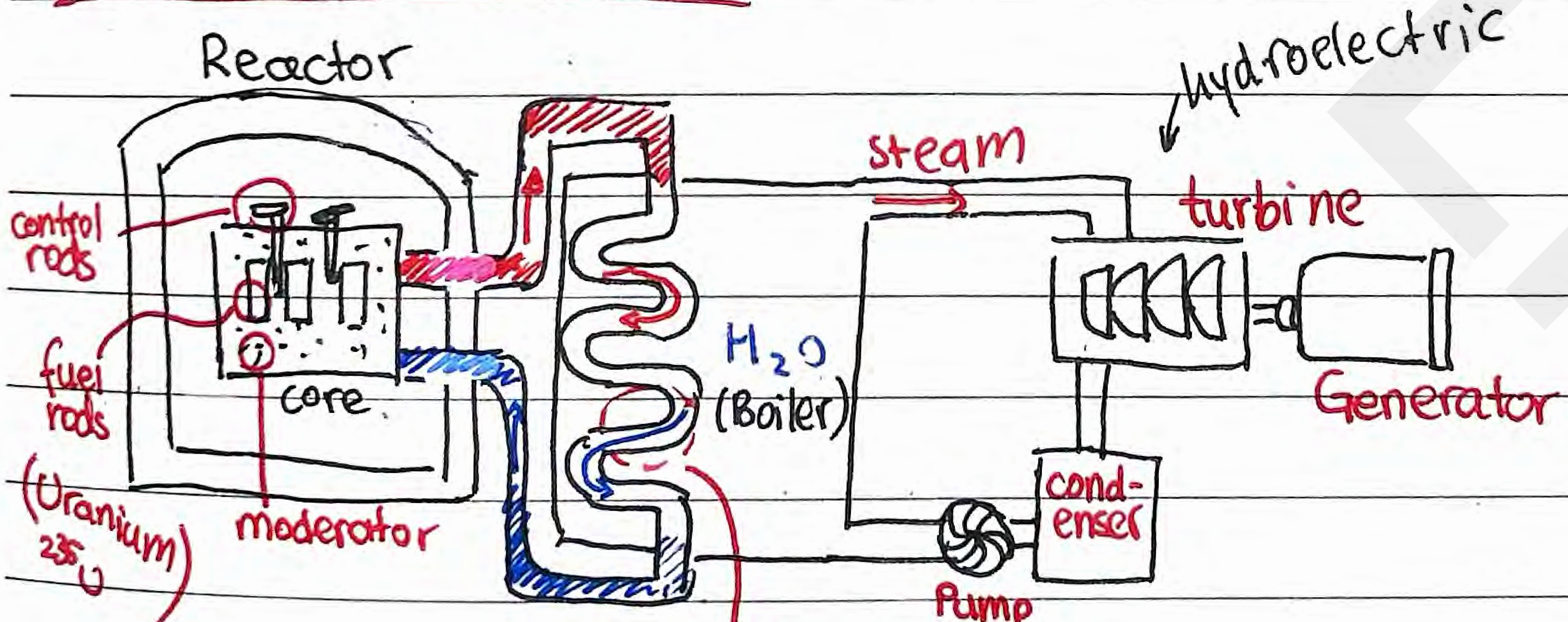
- So this means ratio between masses is ratio between energies → E.g. Fraction of uranium mass to energy of reaction

Thus $\frac{\Delta m}{M_U} = \frac{\Delta E_{BE}}{\Delta E_{tot}}$

> Spontaneous fission can occur but highly rare

- Induced fission is more common since energy required for reaction to start (even if it's exothermic)

Nuclear Power Plant



Moderator – Slows down the speed of neutrons (made of carbon, water)

Control Rods – Absorbs the neutrons (made of boron)